Charge to Mass Ratio for Electron-Lab Report (By Leon Qian and Terry Chan)

1. Introduction: This experiment measures the charge-to-mass ratio (e/m) of the electron by observing its motion in a magnetic field. Electrons are accelerated through a known voltage and bent into a circular path using Helmholtz coils. By measuring the radius of the path and using the applied voltage and current, we can calculate the charge-to-mass ratio. This setup helps demonstrate the fundamental relationship between electric and magnetic fields and the behavior of charged particles.

2. Procedure: Begin by assembling the apparatus according to the provided diagrams. Power on the filament of the electron gun and allow it to heat up for sufficient time. Gradually adjust the input voltage and current until the electron beam becomes visible and forms a well-defined, fully enclosed circular path. Adjust the bulb position so that the beam maintains a circular orbit. Conduct two sets of measurements: one with a constant accelerating voltage and the other with a constant coil current. Record the readings from the voltmeter, ammeter, and measure the electron beam's diameter using the self-illuminated scale. Ensure all measurements are taken with their respective uncertainties. Finally, analyze the data and plot the relevant graphs using Python.

3. Method:
    - Measuring the diameter of the beam path: Despite turning off all the lights in the lab, the self-illuminated scale was still not clearly visible for accurate readings. To overcome this, we used our phones to take high-exposure pictures. We ensured the phone was positioned as horizontally as possible to the scale and turned the exposure to maximum. Each time a measurement was taken, we captured a photo of the scale. These high-exposure images made the scale much more visible, allowing us to read the measurements accurately from the photos. To reduce parallax errors during the measurement electron beam path diameter, we utilized a plastic reflector. The reflector helped to align our line of sight with the scale by providing a clear reflection of the beam path and the scale. This setup ensured that the scale was viewed directly from the same angle each time, minimizing

discrepancies that could arise from viewing the scale from different angles. Additionally, by carefully positioning the camera at the same horizontal level as the scale, most of the parallax errors were eliminated, leading to more accurate measurements.

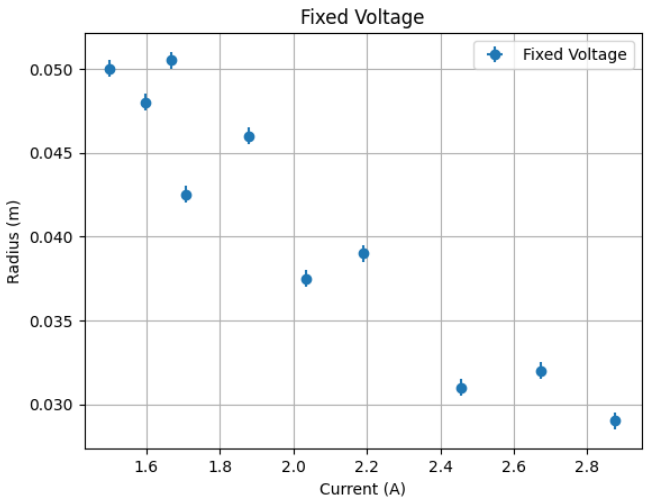
- Measuring the voltage, current, and their uncertainties: The current, voltage, and their respective uncertainties were displayed on the multimeter. To address the fluctuations in the readings, we used the multimeter's average measurement function. We captured a picture of the multimeter display, including both the measurement and the uncertainty, at the same time as taking a photo of the beam diameter measurement. This approach ensured that we synchronized the readings and uncertainties for maximum precision. By reviewing the captured data, we could accurately record both the values and the associated uncertainties.

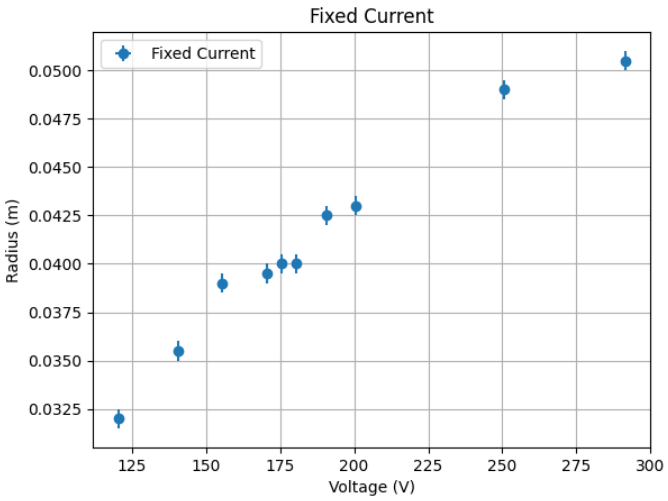
4. Theory: This experiment measures the charge-to-mass ratio (e/m) of the electron by analyzing its motion in a magnetic field.
    - The force on the electron is given by the Lorentz force: $\vec{F} = e\vec{v} \times \vec{B}$.
    - The magnetic force causes the electron to move in a circular path, with the relationship: $evB = m\frac{v^2}{r}$, where r is the radius of the path, v is the electron's velocity, and B is the magnetic field.
    - The velocity is related to the accelerating potential ΔV by: $e\Delta V = \frac{1}{2}\, mv^2$.
    - The combined expression for the charge-to-mass ratio is given by:

$$\frac{1}{r} = \sqrt{\frac{e}{2m}}\frac{B}{\sqrt{\Delta V}}$$

where B is the strength of the magnetic field the electron is in.

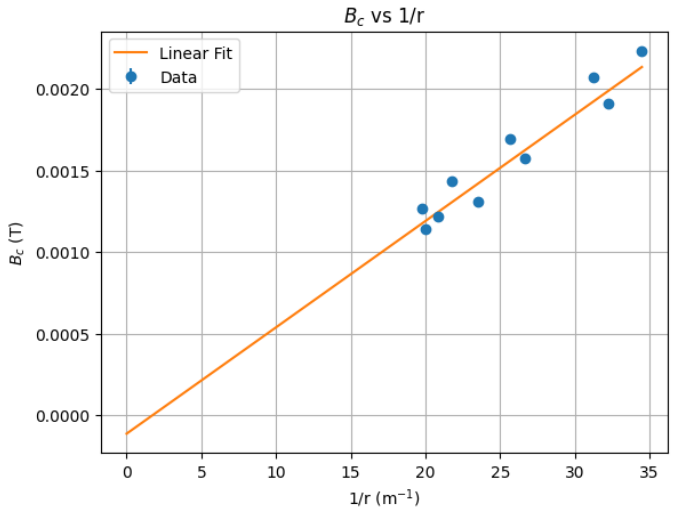
5. Results (Graphs and Interpretations):

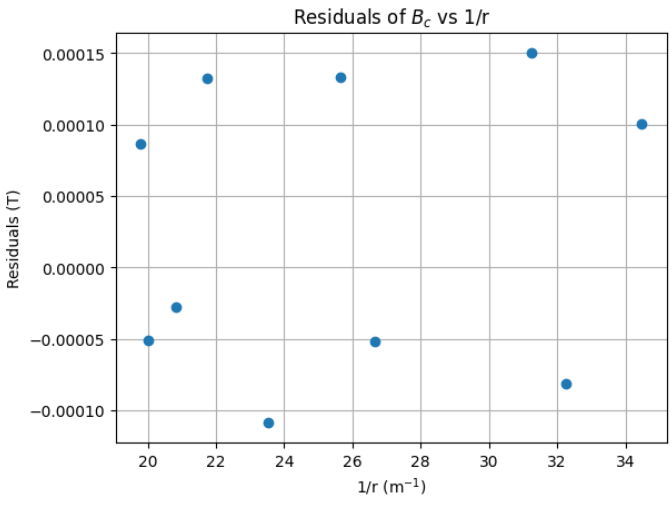




Left graph: Plot of the radius of the electron beam path against the current through the Helmholtz coils, with a constant accelerating voltage (As the current increases, the radius decreases, indicating a stronger magnetic force is acting on the electrons).

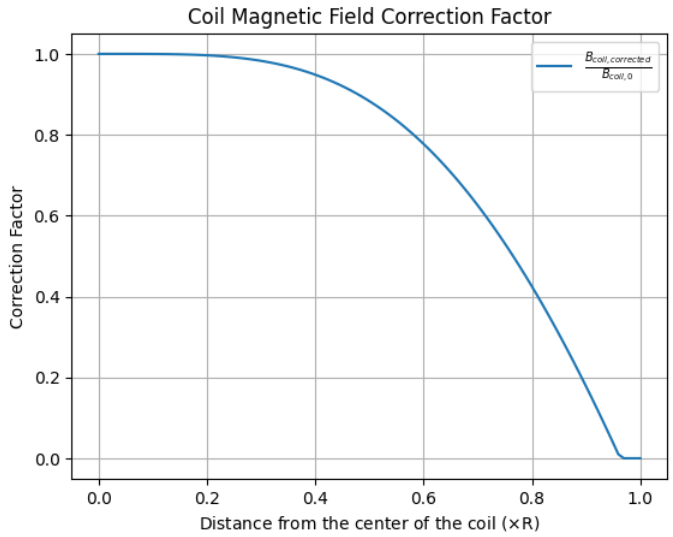
Right graph: Plot of the radius of the electron beam path against the accelerating voltage, with a constant current through the Helmholtz coils (As the voltage increases, the radius increases, showing the electrons gain more kinetic energy).



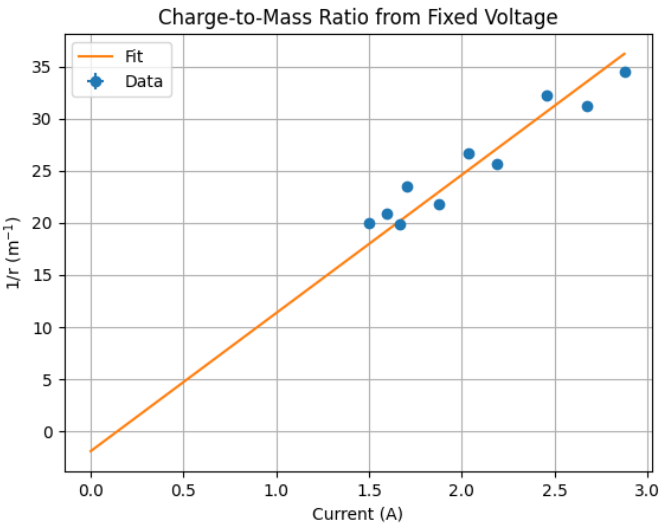


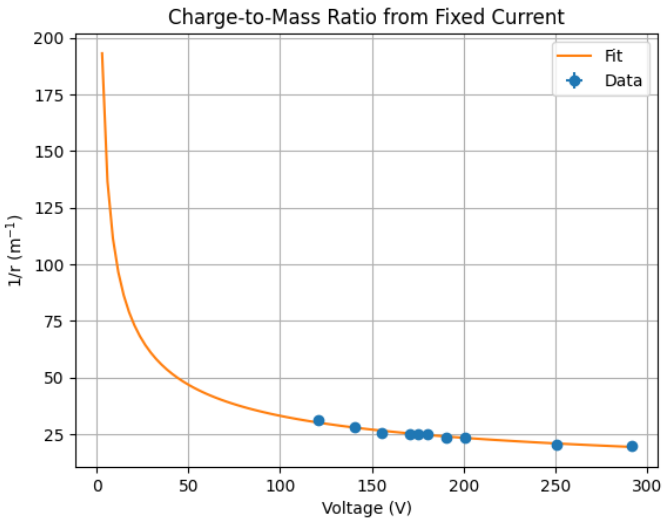
Left graph: Plot of the coil-generated magnetic field as a function of the inverse radius of the electron path (The linear fit demonstrates a reciprocal relationship between the magnetic field and the radius, consistent with the theoretical predictions for the charge-to-mass ratio experiment).

Right graph: Plot of the residuals of the linear fit for the magnetic field and the inverse radius of the electron path (The residuals plot shows minimal deviations, indicating the quality of the fit. The relatively small residuals suggest that the model fits the data well, with no significant outliers).



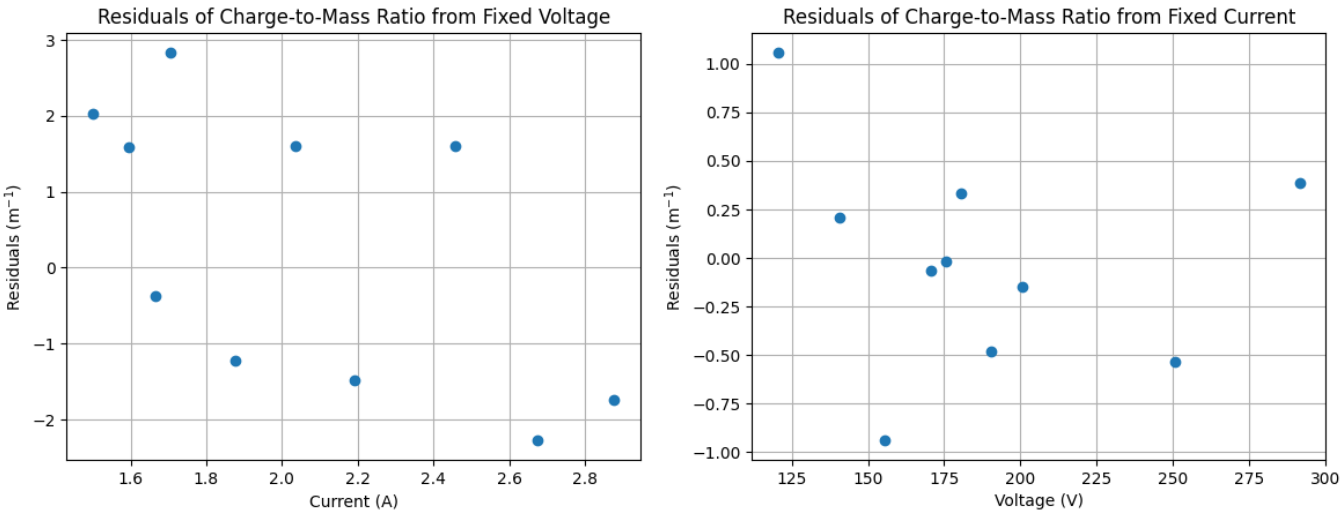
*Plot of the coil magnetic field correction factor as a function of the distance from the center of the coil, expressed as a fraction of the coil radius R. The correction factor decreases as the distance from the center increases, illustrating the variation of the magnetic field strength as the electrons move further from the central axis of the coil. Since the function eventually becomes negative, it was clamped to zero for negative values.





Left graph: Plot of $\frac{1}{r}$ versus current for a fixed accelerating voltage (The linear relationship between $\frac{1}{r}$ and the current indicates the expected proportionality, confirming the theory).

Right graph: Plot of $\frac{1}{r}$ versus voltage for a fixed current (The curve follows the theoretical inverse square root relationship between $\frac{1}{r}$ and the voltage, consistent with the predictions for the charge-to-mass ratio experiment).



Left graph: Residuals of the fit for the fixed voltage data. The random scatter of the residuals shows the model captures most non-linear relationships in the data. The proximity of the residuals to zero also indicates a good fit.

Right graph: Residuals of the fit for the fixed current data. The residuals highlight the differences between the measured values and the model (The relatively small residuals indicate a good fit, with no significant outliers).

6. *Correction Factor Discussion: In the analysis of our data, we found that the external magnetic field was approximately six times stronger than the geomagnetic field and was oriented in the opposite direction. This unexpected result could be due to interference from nearby devices, such as magnets in electronic devices like watches, phones, and laptops.

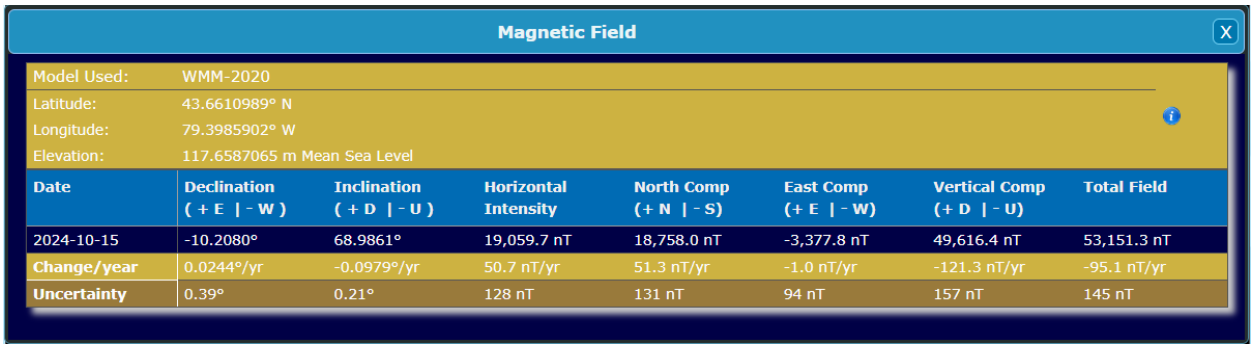
7. Influence of Nearby Ferromagnetic Materials on the Electron Trajectory: During the experiment, we investigated the effect of nearby ferromagnetic materials, specifically by bringing a cell phone close to the glass bulb containing the electron beam. When the phone screen was positioned near the bulb, we observed a noticeable distortion in the electron beam trajectory, where the beam appeared to bend (spirals outwards) in the direction opposite to the phone screen. This effect suggests that the magnetic field produced by the phone's internal components, such as magnets or electronic circuits, can significantly interact with the field generated by the Helmholtz coils. This interaction can introduce errors in the measurement of the electron trajectory, particularly by altering the radius of the beam's circular path. The distortion demonstrates that external magnetic field is enough to affect the precise measurements required for calculating the charge-to-mass ratio.

8. Value Outputs from Python Analysis (Charge-to-Mass Ratio):
   e/m from fixed voltage = $(2.128636 \pm 0.000026) \times 10^{11}$ C/kg
   e/m from fixed current = $(2.465366 \pm 0.000031) \times 10^{11}$ C/kg
   e/m from fixed voltage / e/m from fixed current = 0.863416 ± 0.000015
   Actual value of e/m = $(1.75882000838 \pm 0.00000000055) \times 10^{11}$ C/kg

9. Conclusion: This experiment measured the charge-to-mass ratio of the electron by observing its motion in a magnetic field. The results closely matched the theoretical predictions, the results demonstrated the relationship between electric and magnetic fields on charged particles and highlighted the need to control external influences for accurate measurements.

Appendix

The geomagnetic data was taken from the magnetic field calculator at [NCEI Geomagnetic Calculators]. The coordinates and altitude were found using Google Earth ([Google Earth]).



Magnetic Field

Model Used:	WMM-2020
Latitude:	43.6610989° N
Longitude:	79.3985902° W
Elevation:	117.6587065 m Mean Sea Level

Date	Declination (+E | -W)	Inclination (+D | -U)	Horizontal Intensity	North Comp (+N | -S)	East Comp (+E | -W)	Vertical Comp (+D | -U)	Total Field
2024-10-15	-10.2080°	68.9861°	19,059.7 nT	18,758.0 nT	-3,377.8 nT	49,616.4 nT	53,151.3 nT
Change/year	0.0244°/yr	-0.0979°/yr	50.7 nT/yr	51.3 nT/yr	-1.0 nT/yr	-121.3 nT/yr	-95.1 nT/yr
Uncertainty	0.39°	0.21°	128 nT	131 nT	94 nT	157 nT	145 nT



Calculate Magnetic Field

Latitude: 43.6610989 ● S ○ N

Longitude: 79.3985902 ○ W ● E

Elevation: ● GPS ○ Mean sea level

117.6587065 Meters

Model: ○ WMM (2019-2024) ● IGRF (1590-2024) ● EMM (2000-2019)

Start Date: Year 2024 Month 10 Day 15

End Date: Year 2024 Month 10 Day 15

Step size: 1.0

Result format: ● HTML ● XML ○ CSV ● JSON

Calculate